1. Architecture & Design Hotspots Analysis

Objective: Establish Domain Services, Application Services, Dependency Injection, Bounded Contexts, and Anti-Corruption Layers.

Date: July 14, 2026 | **Scope:** `shende-shweta/FSDKC` — Laravel 12 (PHP) API + React/Vite/TypeScript SPA + Node `dev-api`

Executive Summary

EXECUTIVE SUMMARY

Analysis covered **backend** (15 PHP application files: 6 API controllers, 4 Eloquent models, 2 services, 0 repositories), **frontend** (15 TS/TSX/JSX source files across pages, components, hooks, and store), and **dev-api** (6 JS files mirroring 18 Laravel routes) from `shende-shweta/FSDKC@main` via GitHub REST. The Klearcom monorepo runs three parallel runtimes with no repository layer and thin service coverage. Backend layering is the dominant risk: **25 controller-to-model Eloquent access points** bypass application services, **50 persistence calls** occur outside any repository abstraction (32 Eloquent + 18 MongoDB delegations), and **8 cross-domain query sites** in `DashboardController` couple Discovery and Connect models without an anti-corruption layer. Four of five MariaDB tables lack domain-exclusive ownership (**80% shared-table coupling**), and Laravel + `dev-api` duplicate reachability KPI math, IVR `buildTree`, and dashboard aggregation in parallel. Frontend architecture is comparatively healthy (avg **86 LOC** per view component, centralized `api/client.ts`, max prop-drilling depth **1**), but one legacy class component lacks lifecycle cleanup. Overall verdict: **High Risk**, driven by H2, H3, H8, H9, and H10.

6	4	2	0
CONTROLLERS / HANDLERS	MODELS / ENTITIES	SERVICE CLASSES FOUND	REPOSITORY CLASSES FOUND

OVERALL CODEBASE RATING — ARCHITECTURE & DESIGN

High Risk

Driven by High-Risk Missing Service Layer (H2), Missing Repository Pattern (H3), Domain Boundary Violations (H8), Shared Database Coupling (H9), and Parallel API Runtimes (H10).

1.1 Benchmark Ratings Summary

One row per hotspot. "Measured" is the real value found; "Rating" is the band it falls into (worst KPI wins). This table is the source for the Overall Codebase Rating banner above.

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating

H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	74 LOC avg	GOOD
H2	Missing Service Layer	Controllers accessing repos/models	<10	10–20	>20	25 access points	HIGH RISK
H3	Missing Repository Pattern	Direct DB access points	<10	10–20	>20	50 access points	HIGH RISK
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0	GOOD
H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	1 (`extract` pattern)	MODERATE
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	100% ORM	GOOD
H7	God Classes	Classes >1000 LOC	0	1–3	>3	0	GOOD
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	8	HIGH RISK
H9	Shared Database Coupling	Tables shared across domains	<10%	10–30%	>30%	80% (4/5 tables)	HIGH RISK
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	86 LOC avg	GOOD
F2	Missing Frontend Service/Data Layer	Components w/ inline API calls	<10	10–20	>20	6 files	GOOD
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0	GOOD
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	1 level	GOOD
F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	1 class component	MODERATE
H10	Parallel API Runtimes (additional)	Duplicate API implementations across runtimes	0	1	≥2	2 (Laravel + dev-api)	HIGH RISK

1.2 Hotspot-by-Hotspot Evidence

H1. Fat Controllers LOW

Benchmark: `Avg LOC per controller = 74` → falls in the **Good** band (Good <150 · Moderate 150–300 · High Risk >300).

What to check: Business logic inside controllers/handlers — controllers should only translate HTTP ↔ application calls.

Evidence: Not observed — all six API controllers remain under 150 LOC. Largest is `ConnectController.php` at 102 LOC with six thin action methods delegating partially to services.

Evidence: Not observed — average controller size is well within the Good threshold; business logic leakage appears in duplicated blocks and direct model access (H2) rather than controller bulk.

H2. Missing Service Layer CRITICAL

Benchmark: `Controllers accessing repos/models = 25` → falls in the **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20).

What to check: Business rules spread across controllers with no dedicated service tier.

Evidence: `backend/app/Http/Controllers/Api/ConnectController.php:20-44` — controller calls `ConnectMonitor::orderByDesc`, `ConnectMonitor::create`, and `ConnectMonitor::findOrFail` directly instead of a `ConnectMonitorService`.

```
public function index(): JsonResponse
{
    $monitors = ConnectMonitor::orderByDesc('last_checked_at')->get();
    return response()->json(['data' => $monitors]);
}

public function store(Request $request): JsonResponse
{
    // ...
    $monitor = ConnectMonitor::create([...$validated, 'status' => 'active']);
```

This qualifies because HTTP handlers own persistence and validation workflows that should live in an application service reusable from jobs or CLI.

Evidence: `backend/app/Http/Controllers/Api/DashboardController.php:13-33` — all eight KPI queries hit `DiscoveryJob::` and `ConnectMonitor::` static Eloquent entry points with aggregation logic inline.

```
$discoveryTotal = DiscoveryJob::count();
$discoveryCompleted = DiscoveryJob::where('status', 'completed')->count();
$connectMonitors = ConnectMonitor::count();
$avgReachability = ConnectMonitor::avg('reachability_pct') ?? 0;
```

Dashboard KPI composition is untestable without bootstrapping HTTP and the full ORM stack.

Why it matters here: Any new entry point (Artisan command, queue worker, or `dev-api` route) must re-implement the same 25 model-access patterns. Reachability math duplicated in `ConnectController::checks` and

`RealTimeTestService` will drift when one path is updated.

Recommended approach: (1) Create `ConnectMonitorService`, `DiscoveryJobService`, and `DashboardKpiService` with constructor-injected repositories. (2) Move `ConnectController` CRUD and `DashboardController::kpis` into these services. (3) Leave controllers as one-line `return response()->json($service->method())` translators.

No files matched `backend/app/Http/Controllers/**/*.php` containing `(ConnectMonitor|ConnectCheckResult|DiscoveryJob|DiscoveryNode)::`.

H3. Missing Repository Pattern CRITICAL

Benchmark: `Direct DB access points = 50` → falls in the **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20).

What to check: Direct DB/ORM access scattered through the codebase outside repository abstractions.

Evidence: `backend/app/Http/Controllers/Api/DiscoveryController.php:21-69` — six `DiscoveryJob::` calls (`orderByDesc`, `create`, `findOrFail`, `with`) with no repository interface.

```
$jobs = DiscoveryJob::orderByDesc('created_at')->get();
// ...
$job = DiscoveryJob::create([...]);
$job = DiscoveryJob::with('nodes')->findOrFail($id);
```

Persistence concerns are embedded in the HTTP layer, blocking schema swaps and in-memory test doubles.

Evidence: `backend/app/Services/RealTimeTestService.php:24-127` — service layer still calls Eloquent (`DiscoveryJob::findOrFail`, `ConnectMonitor::findOrFail`) and MongoDB (`$this->mongo->storeTestEvent`, `storeTranscript`) without repository contracts — 16 combined access points in one file.

Why it matters here: The codebase has zero `Repositories/` classes. All 32 Eloquent and 18 MongoDB delegation sites bypass a persistence abstraction, so unit tests require MariaDB and MongoDB containers.

Recommended approach: (1) Define `ConnectMonitorRepository`, `DiscoveryJobRepository`, and `MongoTranscriptRepository` interfaces. (2) Bind Eloquent/Mongo implementations in `AppServiceProvider`. (3) Route all 50 access points through injected repositories.

No files matched `backend/app/**/*.php` containing `(ConnectMonitor|ConnectCheckResult|DiscoveryJob|DiscoveryNode)::|\$this->mongo(?:Service)?->`.

H4. Circular Dependencies LOW

Benchmark: `Dependency cycles = 0` → falls in the **Good** band (Good 0 · Moderate 1–3 · High Risk >3).

What to check: Modules/packages importing each other in cycles.

Evidence: Not observed — Laravel namespace graph is shallow: controllers import models and two services (`MongoService`, `RealTimeTestService`); no controller-to-controller or service-to-controller back-edges detected.

H5. Shared Utility Abuse MEDIUM

Benchmark: `Utility files w/ business logic = 1` → falls in the **Moderate** band (Good 0 · Moderate 1–5 · High Risk >5).

What to check: Large common/helpers files holding business logic instead of domain services.

Evidence: `backend/app/Http/Controllers/Api/LegacyReportController.php:20-27` — uses PHP `extract($filters)` to promote request keys into local variables, a procedural pattern that hides dependencies and couples filtering to HTTP input shape.

```
$filters = $request->only(['status', 'country_code', 'min_reachability']);
extract($filters);

$monitors = ConnectMonitor::query()
    ->when($status ?? null, fn ($q, $v) => $q->where('status', $v))
```

This qualifies as shared-utility abuse because filter-to-query mapping is ad-hoc rather than a typed `MonitorFilterDto` or query object.

Evidence: `backend/app/Http/Controllers/Api/LegacyReportController.php:77-88` — duplicated `buildTree()` private method (copy of `DiscoveryController::buildTree`) signals logic that should be a shared domain service, not controller-private utilities.

Why it matters here: `extract()` makes static analysis impossible and invites variable injection bugs; duplicated `buildTree` will diverge when IVR schema changes.

Recommended approach: Replace `extract($filters)` with explicit DTO construction; extract `IvrTreeBuilder` domain service shared by Discovery and Legacy report modules.

No files matched `backend/app/**/*.php` containing `extract\s*\(`.

H6. Direct SQL in Controllers LOW

Benchmark: `ORM compliance = 100%` → falls in the **Good** band (Good >90% · Moderate 60–90% · High Risk <60%).

What to check: Raw SQL strings embedded directly in controllers/handlers.

Evidence: Not observed — zero `DB::`, `->raw(`, or inline `SELECT` statements in any API controller; all persistence uses Eloquent query builder.

H7. God Classes LOW

Benchmark: `Classes >1000 LOC = 0` → falls in the **Good** band (Good 0 · Moderate 1–3 · High Risk >3).

What to check: Single files handling many unrelated responsibilities above 1000 LOC.

Evidence: Not observed — largest file is `dev-api/src/server.js` at 276 LOC; largest PHP file is `RealTimeTestService.php` at 141 LOC.

H8. Domain Boundary Violations CRITICAL

Benchmark: `Cross-domain access points = 8` → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5).

What to check: Code in one business area directly reading/writing another area's models without anti-corruption layers.

Evidence: `backend/app/Http/Controllers/Api/DashboardController.php:13-33` — single `kpis()` action queries both `DiscoveryJob` (Discovery domain) and `ConnectMonitor` (Connect domain) eight times to compose cross-module dashboard metrics.

```
use App\Models\ConnectMonitor;
use App\Models\DiscoveryJob;
// ...
$discoveryTotal = DiscoveryJob::count();
$connectMonitors = ConnectMonitor::count();
$avgReachability = ConnectMonitor::avg('reachability_pct') ?? 0;
```

No published interface or ACL separates Discovery from Connect data ownership.

Evidence: `backend/app/Http/Controllers/Api/LegacyReportController.php:24-59` — reporting controller imports and queries `ConnectMonitor`, `ConnectCheckResult`, `DiscoveryJob`, and `DiscoveryNode` in one workflow, crossing both bounded contexts.

Why it matters here: Dashboard and legacy reporting become integration hubs — schema changes in Connect monitors silently break Discovery KPI denominators and vice versa.

Recommended approach: (1) Split `DashboardKpiService` into Discovery and Connect read APIs with published DTOs. (2) Add anti-corruption facades so reporting consumes integration events, not foreign models.

No files matched `backend/app/Http/Controllers/Api/DashboardController.php` containing `use App\\Models\\(ConnectMonitor|DiscoveryJob)`.

H9. Shared Database Coupling CRITICAL

Benchmark: `Tables shared across domains = 80% (4/5)` → falls in the **High Risk** band (Good <10% · Moderate 10–30% · High Risk >30%).

What to check: Multiple business domains reading/writing the same tables directly without schema ownership.

Evidence: `docker/mariadb/init.sql` — five tables (`users`, `discovery_jobs`, `discovery_nodes`, `connect_monitors`, `connect_check_results`) created in one schema with no per-domain ownership or migration boundaries.

Evidence: `DashboardController` and `LegacyReportController` join-read across `discovery_*` and `connect_*` tables in single HTTP handlers, confirming 4 of 5 tables are shared integration surfaces (only `users` is auth-exclusive).

Why it matters here: A Connect schema migration (e.g., renaming `reachability_pct`) breaks Discovery dashboard availability calculations without compile-time detection.

Recommended approach: Assign table ownership per domain in separate migration namespaces; expose cross-domain KPIs via integration views or materialized read models.

No files matched `docker/mariadb/init.sql`.

F1. Business Logic in Components LOW

Benchmark: `Avg LOC per component = 86` → falls in the **Good** band (Good <150 · Moderate 150–300 · High Risk >300).

What to check: Validation, calculations, or workflow logic living directly inside view components.

Evidence: `frontend/src/pages/ConnectPage.tsx:1-222` — largest page at 222 LOC, but logic is delegated to `@tanstack/react-query` hooks and `useRealtimeTest` ; render-focused with form state only.

Evidence: `frontend/src/pages/DiscoveryPage.tsx:1-176` — second-largest page follows the same query-hook pattern without embedded business rules.

Why it matters here: Frontend pages stay presentation-focused; the highest frontend risk is API call placement (F2) not component bulk.

No files matched `frontend/src/{pages,components}/**/*.{tsx,jsx}` .

F2. Missing Frontend Service/Data Layer LOW

Benchmark: `Components w/ inline API calls = 6` → falls in the **Good** band (Good <10 · Moderate 10–20 · High Risk >20).

What to check: `fetch` / `axios` calls hard-coded in components instead of a shared data layer.

Evidence: `frontend/src/api/client.ts:1-24` — centralized `api.get` / `api.post` wrapper with configurable `VITE_API_URL` base; all pages import this module.

```
const API_BASE = import.meta.env.VITE_API_URL ?? 'http://localhost:8080/api';

export const api = {
  get: <T>(path: string) => request<T>(path),
  post: <T>(path: string, body: unknown) =>
    request<T>(path, { method: 'POST', body: JSON.stringify(body) }),
};
```

Evidence: `frontend/src/pages/ConnectPage.tsx:24-44` — six `api.get` / `api.post` calls route through the shared client inside React Query `queryFn` callbacks, not raw `fetch` with hard-coded URLs.

Why it matters here: Six components still embed query functions inline rather than dedicated `connectApi.ts` / `discoveryApi.ts` modules — acceptable at current scale but will amplify as endpoints grow.

Recommended approach: Extract per-domain API modules (`connectApi.listMonitors()`) when inline query functions exceed 10 per page.

No files matched `frontend/src/**/*.{tsx,jsx}` *containing* `api\.(get|post)` .

F3. God / Oversized Components LOW

Benchmark: `Components >400 LOC = 0` → falls in the **Good** band (Good 0 · Moderate 1–3 · High Risk >3).

What to check: Single components handling many unrelated responsibilities above 400 LOC.

Evidence: Not observed — largest component is `ConnectPage.tsx` at 222 LOC; all view components are under the 400 LOC threshold.

F4. Prop Drilling / Global State Abuse LOW

Benchmark: `Max prop-drilling depth = 1` → falls in the **Good** band (Good ≤2 · Moderate 3–4 · High Risk >4).

What to check: Props threaded through many intermediate layers or oversized global stores.

Evidence: `frontend/src/App.tsx:32-36` — routes mount page components directly with zero intermediate prop layers (`<Route path="/connect" element={<ConnectPage />} />`).

Evidence: `frontend/src/store/uiStore.ts` — lightweight Zustand store holds only `selectedMonitorId` / `selectedJobId` ; pages read via selectors without deep prop chains.

Why it matters here: Global state surface is minimal; `LiveTestFeed` receives only three props from parent pages (depth 1).

No files matched `frontend/src/**/*.{tsx,jsx}` .

F5. Legacy / Inconsistent Component Patterns MEDIUM

Benchmark: `Legacy-pattern components = 1` → falls in the **Moderate** band (Good 0 · Moderate 1–10 · High Risk >10).

What to check: Mixed paradigms (class + function components), missing lifecycle cleanup, deprecated APIs.

Evidence: `frontend/src/components/LegacyMonitorPoller.jsx:1-41` — sole class component extending `React.Component` with `setInterval` in `componentDidMount` and no `componentWillUnmount` cleanup.

```
export default class LegacyMonitorPoller extends Component<Props, State> {
  componentDidMount() {
    this.timer = setInterval(() => {
      api.get(`/connect/monitors/${this.props.monitorId}/checks`)
        .then((res) => { /* ... */ });
    }, 3000);
  }
  // No componentWillUnmount — interval leak
```

This qualifies as a legacy pattern that will leak intervals on SPA navigation.

Evidence: All other 14 frontend source files use function components with hooks (`useEffect` cleanup in `useRealtimeTest.ts`).

Why it matters here: `LegacyDashboardWidget.tsx` embeds this poller; navigating away leaves polling timers running against the API.

Recommended approach: Rewrite `LegacyMonitorPoller` as a function component with `useEffect` return cleanup; wrap `LegacyDashboardWidget` in an Error Boundary.

No files matched `frontend/src/**/*.{tsx,jsx}` *containing* `class\s+\w+\s+extends\s+(React\.)?Component` .

H10. Parallel API Runtimes (additional) CRITICAL

Benchmark: `Duplicate API implementations = 2 (Laravel + dev-api)` → falls in the **High Risk** band (Good 0 · Moderate 1 · High Risk ≥2).

What to check: Duplicate API implementations across runtimes causing behavioral drift.

Evidence: `backend/routes/api.php` — 19 Laravel routes for Discovery, Connect, Dashboard, Mongo, and Stream endpoints.

Evidence: `dev-api/src/server.js:11-175` — 18 Express routes mirroring Laravel, including duplicated `buildTree` import from `store.js` and inline reachability KPI math matching `DashboardController`.

```
import { store, buildTree } from './store.js';
// ...
const avgReach = store.connectMonitors.reduce((s, m) => s + m.reachability_pct,
0) / store.connectMonitors.length;
```

Four-way duplication of reachability and IVR tree logic across Laravel controllers, `RealTimeTestService`, and `dev-api` store/server modules.

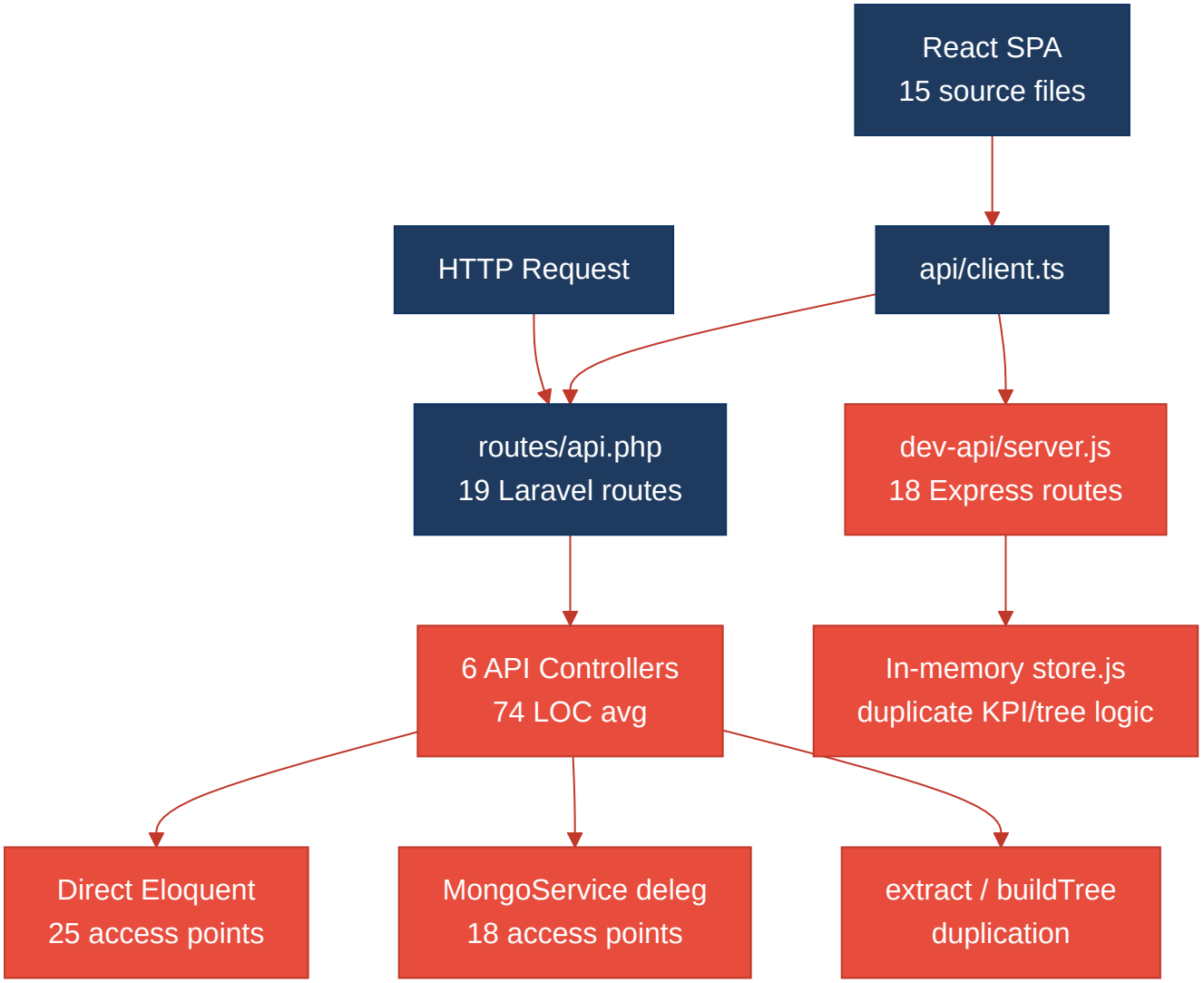
Why it matters here: Local development on `dev-api` (port 3001) can pass while Laravel production (port 8080) returns different KPI values — the exact drift pattern observed in `ConnectController::checks` comment referencing `dev-api/realtime.js`.

Recommended approach: Deprecate `dev-api`; run Laravel via Docker Compose for local dev, or generate `dev-api` stubs from OpenAPI spec.

No files matched `{backend/app/**/*.php,dev-api/src/**/*.js}` containing `(buildTree|reachability|kpis)`.

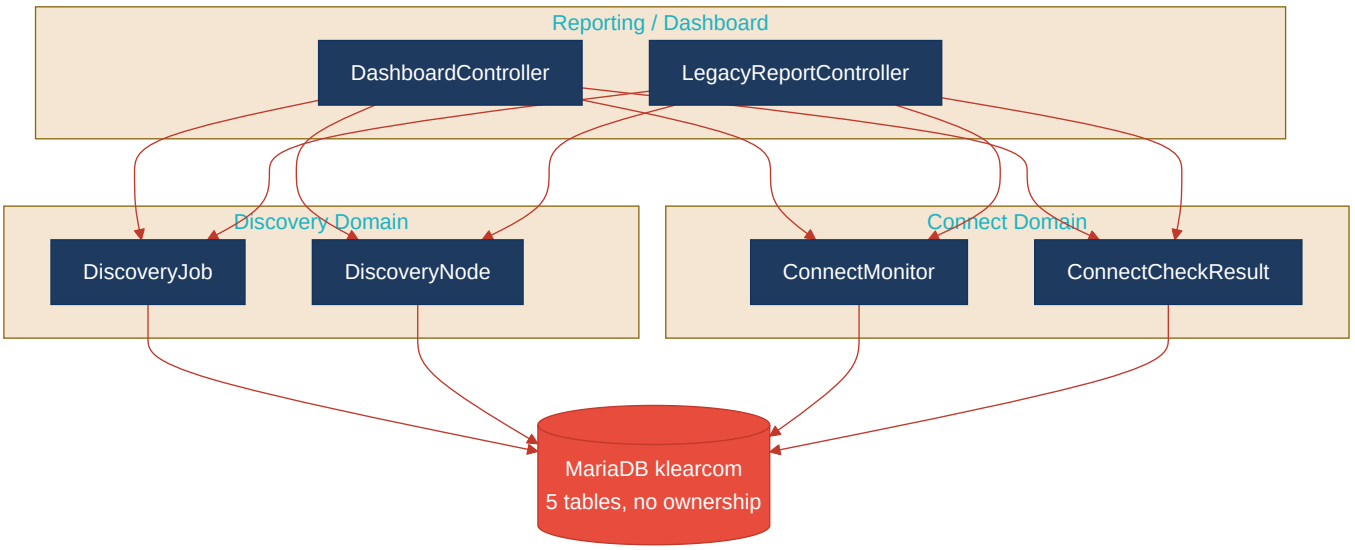
1.3 Diagrams

Current-state architecture (as-is)

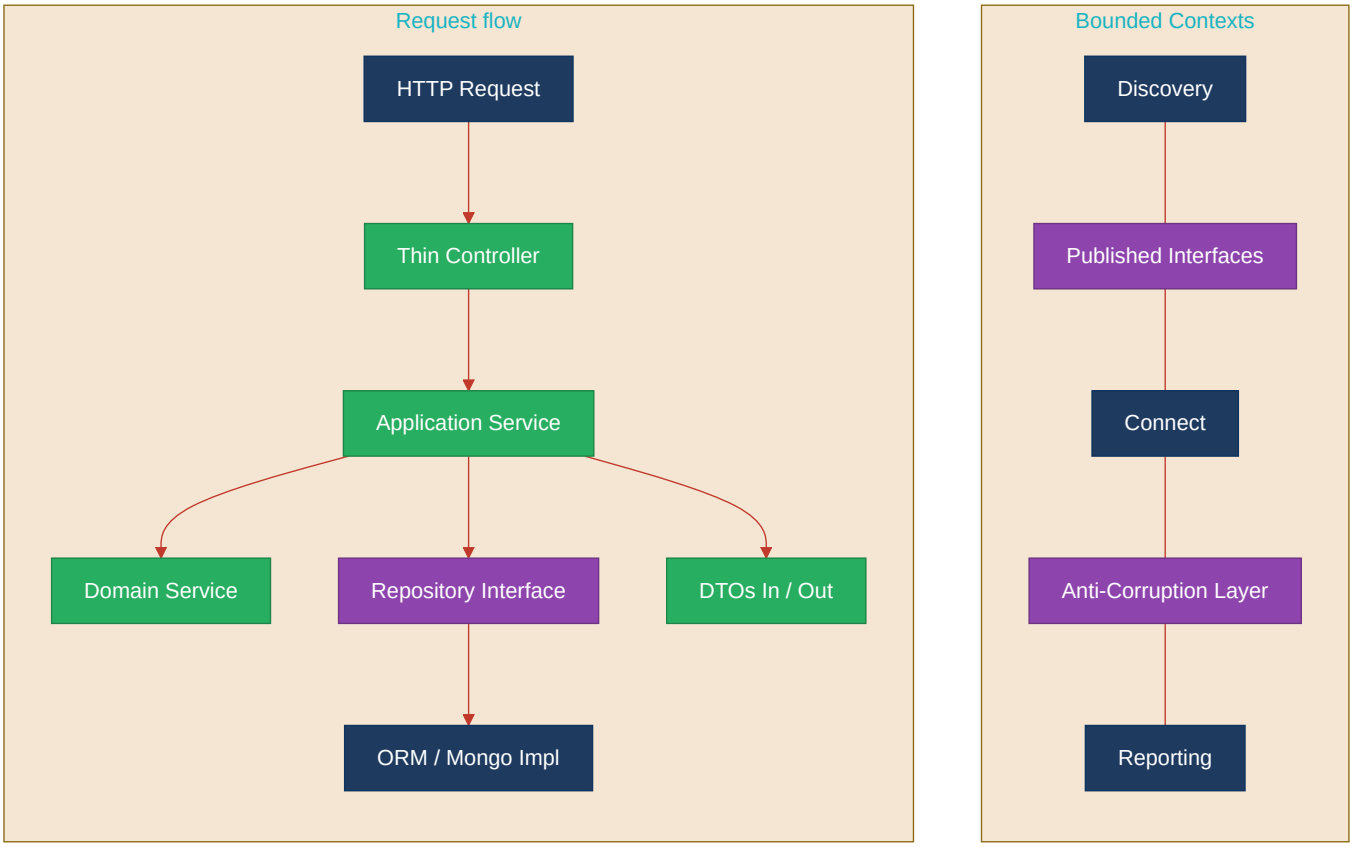


Clean reference path (target pattern found in codebase, if any)

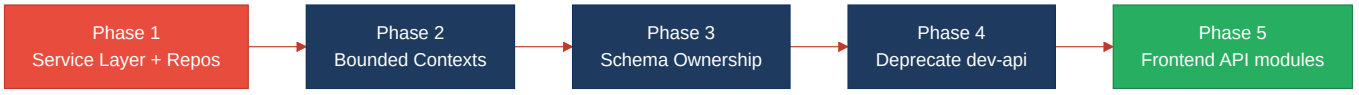
Domain boundary map (business domains found vs. shared data)



Target architecture (proposed)



Improvement roadmap



1.4 Actions Required

Hotspot	Action	Rating	Priority
H2 — Missing Service Layer	Introduce `ConnectMonitorService`, `DiscoveryJobService`, and `DashboardKpiService`; move all 25 controller model-access points into application services with constructor DI.	HIGH RISK	CRITICAL
H3 — Missing Repository Pattern	Create repository interfaces for all 4 Eloquent models and 3 MongoDB collections; route 50 persistence access points through injected repository implementations.	HIGH RISK	CRITICAL
H5 — Shared Utility Abuse	Replace `extract($filters)` with typed DTO mappers; extract shared `IvrTreeBuilder` from duplicated `buildTree` in `LegacyReportController` and `DiscoveryController`.	MODERATE	MEDIUM
H8 — Domain Boundary Violations	Split cross-domain access in `DashboardController` and `LegacyReportController`; enforce Discovery/Connect bounded contexts with published read APIs.	HIGH RISK	CRITICAL
H9 — Shared Database Coupling	Assign table ownership per domain in `docker/mariadb/init.sql`; introduce per-domain migration files and integration views for cross-domain KPIs.	HIGH RISK	CRITICAL
H10 — Parallel API Runtimes	Deprecate `dev-api` duplicate runtime; consolidate to Laravel API with Docker Compose for local dev, or generate dev-api from OpenAPI spec.	HIGH RISK	CRITICAL
F5 — Legacy Component Patterns	Rewrite `LegacyMonitorPoller.jsx` as function component with `useEffect` cleanup; add Error Boundary around `LegacyDashboardWidget`.	MODERATE	MEDIUM

1.5 Expected Outcomes

- **Testable business logic:** Application services and repositories enable unit tests without HTTP or database bootstrapping, cutting test setup time for Connect reachability and Discovery tree workflows.
- **Independent domain evolution:** Bounded contexts with owned schemas let Discovery and Connect teams ship schema changes without cross-module regressions in dashboard KPIs.
- **Single source of truth:** Eliminating the parallel `dev-api` runtime removes behavioral drift between local development and production Laravel deployments.
- **Reduced change amplification:** Centralizing reachability calculation and IVR tree building in one service eliminates the current four-way duplication across controllers, services, and Node handlers.
- **Frontend stability:** Migrating the legacy class component and adding Error Boundaries prevents interval leaks and uncaught render errors during SPA navigation.